Section 13.2: Derivatives and Integrals of Vector Functions

Vector Functions

MTH 310
Calculus III

Derivatives of Vector Functions

Derivative of a Vector Function

$$\frac{d\vec{r}}{dt} = \vec{r}\,'(t) = \lim_{h \to 0} \frac{\vec{r}(t+h) - \vec{r}(t)}{h}$$

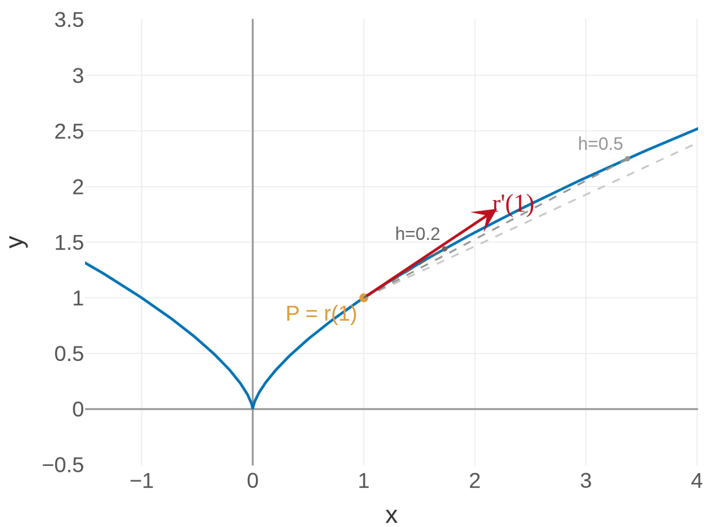
If $\vec{r}(t) = \langle f(t),\, g(t),\, h(t) \rangle$, then:

$$\vec{r}\,'(t) = \langle f'(t),\, g'(t),\, h'(t) \rangle$$

Just **differentiate each component** separately.

Derivative = velocity. The computation is simple: differentiate component by component, just like Calc I.

What Does $\vec{r}\,'(t)$ Mean Geometrically?



Interactive version

$\vec{r}\,'(t)$ is the **velocity vector** — direction of travel + speed, all in one package.

The Unit Tangent Vector

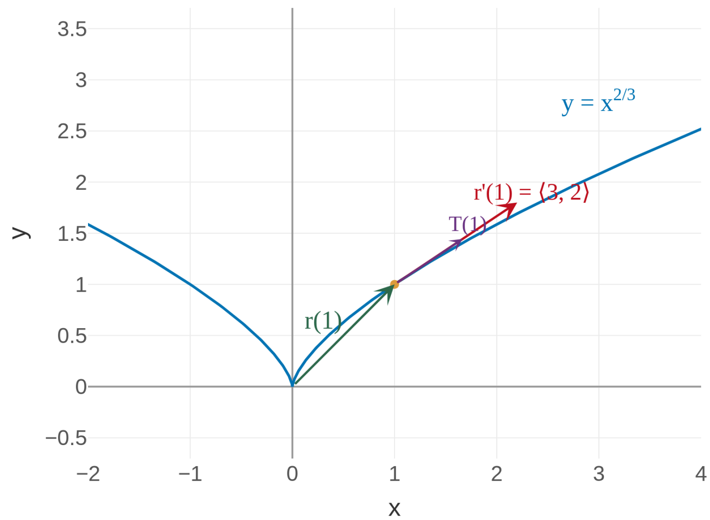
Unit Tangent Vector

$$\vec{T}(t) = \frac{\vec{r}\,'(t)}{|\vec{r}\,'(t)|}$$

This is a unit vector ($|\vec{T}| = 1$) that points in the direction of motion.

Dividing by the magnitude strips away speed and isolates pure **direction**.

Example 1: Find $\vec{r}\,'(t)$ and $\vec{T}(t)$ for $\vec{r}(t) = \langle t^3,\, t^2 \rangle$ at $t = 1$

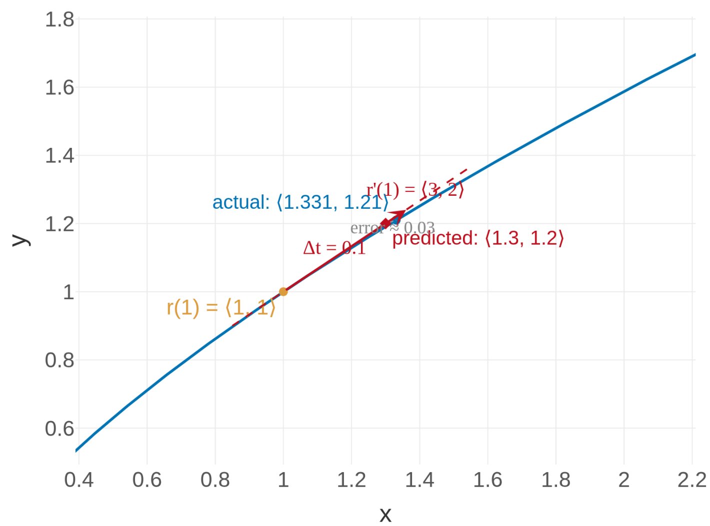


Interactive version

At $(1,1)$, the particle is moving in the direction $\langle 3,2\rangle$ with speed $\sqrt{13} \approx 3.6$.

The Velocity Vector as a Prediction

$\langle 1.331,\ 1.21\rangle$ vs. $\langle 1.3,\ 1.2\rangle$ — almost identical! The velocity vector is the **linear approximation** of motion along the curve.



Interactive version

Your Turn: Find $\vec{r}\,'(t)$ and $\vec{T}(t)$ for $\vec{r}(t) = \langle \cos t,\ \sin t,\ t\rangle$ at $t = 0$

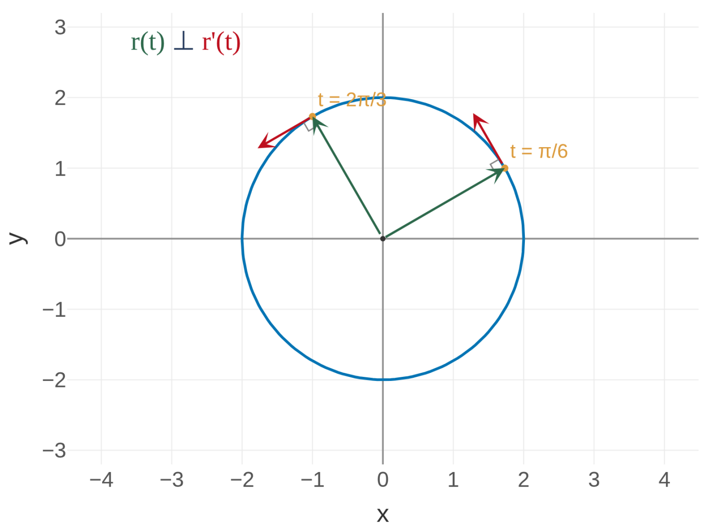
Try it before looking at the solution.

Theorem: Constant Distance $\Rightarrow$ Orthogonality

Orthogonality Theorem

If $|\vec{r}(t)| = c$ (constant), then $\vec{r}(t) \cdot \vec{r}\,'(t) = 0$.

In other words: **position** $\perp$ **velocity**.



Interactive version

On a sphere or circle, velocity is always **perpendicular to the radius**. This is why orbits are tangent to their paths.

Integrals of Vector Functions

Definite Integral of a Vector Function

$$\int_a^b \vec{r}(t)\,dt = \left\langle \int_a^b f(t)\,dt,\ \int_a^b g(t)\,dt,\ \int_a^b h(t)\,dt \right\rangle$$

No new techniques — just Calc I integration applied to each component. The only wrinkle: $\vec{C}$ is a constant **vector**, not a number.

Example 2: Evaluate $\displaystyle\int_1^4 \left(2t^{3/2}\,\mathbf{i} + \frac{1}{1+t^2}\,\mathbf{j} + (t+1)\sqrt{t}\,\mathbf{k}\right) dt$

Result: $\frac{124}{5}\,\mathbf{i} + \left(\arctan 4 - \frac{\pi}{4}\right)\mathbf{j} + \frac{256}{15}\,\mathbf{k}$

Your Turn: Evaluate $\int_0^{\pi/2} \langle \cos t,\ \sin t,\ 1\rangle\, dt$

Try it before looking at the solution.

Homework

Section 13.2: 3, 5, 7, 9, 13, 19, 27, 29, 37, 39, 41